

5. ADSORPTION

Adsorption is a surface separation process involving the selective transfer of solutes (*adsorbates*) in a fluid phase to (and from) the surface of a solid (*the adsorbent*).

The affinity of a solute for a particular adsorbent depends on molecular characteristics such as size, shape, polarity, partial pressure or concentration of the solute and temperature.

- Adsorbents are normally highly porous solids (to create a large surface area for adsorption per unit volume) and have specific surface areas as high as 2000 m^2/g. Zeolites, MOFs
- Adsorbent particles are normally spherical beads with sizes ranging from a few μm to a few mm.
- The adsorbent beads are normally packed into a column creating a fixed bed of adsorbent, through which fluids are pumped.

Adsorption can be classified as:

- **Physisorption**: the interaction between adsorbate and adsorbent is caused by van der Waals forces and it is essentially physical by nature. Bonding energies range from 10 to 70 kJ/mol. Forces are not too strong so adsorption may be easily reversed.
- **Chemisorption**: the interaction between the adsorbate and adsorbent is of a chemical nature involving the exchange and sharing of electrons, or possibly molecules forming atoms or radical. Chemisorption is restricted to one layer of molecules on the surface. Covalent bond energies range from 200 to 500 kJ/mol. Chemisorption is usually difficult to reverse and in some cases is considered irreversible.

Adsorption is always accompanied by liberation of heat. For physical adsorption, the amount of heat is similar in magnitude to the heat of condensation. For chemisorption is greater and of an order of magnitude normally associated with a chemical reaction.

Steps in an adsorption cycle

The separation of adsorbates via adsorption is intrinsically a non-continuous process, normally carried out in cycles with well-distinguished steps:

1. Adsorption or Binding: where the fluid is contacted with the adsorbent and it binds to its surface.

2. Washing: adsorbent is washed to remove unbound and weakly bound compounds. This step may result in a small loss of adsorbed adsorbent due to the reversible nature of adsorption.
3. Elution or Desorption: where the adsorbate is recovered from the adsorbent. For this, it is necessary to change the chemical or physical characteristics of the fluid phase where the adsorbate no longer has any affinity for the solid adsorbent phase and dissociates from the adsorbent and (re)joins the bulk fluid phase
4. Regeneration: the adsorbent is brought back to its original conditions and it can be re-used.

- In gas adsorption processes, washing does not take place, and the desorption and regeneration steps are usually merged.
- In liquid adsorption processes, in particular those used for the isolation of biomolecules, all steps take place .
    - In the ion exchange adsorption of biomolecules carried out in the liquid phase, the desired adsorbate is bound to the adsorbent via electrostatic interactions resulting from opposite charges on the surfaces of the adsorbate and adsorbent. Adsorbate can be eluted either by changing the pH of the irrigating fluid such that the two species are no longer oppositely charged (thus destroying the attraction between them), or by flowing high concentrations of a simple salt through the bed that effectively compete for the adsorption sites on the adsorbent thus also displacing the biomolecular adsorbate.

5.1 Applications

Vapour phase

- Recovery of evaporated organic solvents – paints, inks manufacture
- Prevention of emissions of volatile organic compounds (VOC) during tank breathing operations (activated carbon)
- Removal of pollutants H_2S, CS_2 and odours from re-circulating air using carbon (e.g. in an aeroplane air conditioning).
- Removal of water from gases using alumina.
- Separation of O_2 and N_2 using molecular sieves such as zeolites (aluminosilicates with a well-defined pore structure and chemical composition).

Liquid Phase

- Removal of organics from aqueous and other solutions, *e.g.* coloured impurities from sugars and food oils with activated carbon.
- Softening and deionisation of water using ion-exchangers, *e.g.* adsorption of Ca^{2+} ions accompanied by desorption of Na^{+} ions (water softening) or
  a sequential two-bed processes involving adsorption of Na^{+} ions, accompanied by desorption of H^{+} ions (yielding an acidic solution), followed by adsorption of Cl^{-} ions adsorbed, accompanied by desorption of OH^{-} ions (to yield a pH neutral solution). (water deionisation)
- Purification of biological macro-molecules, *e.g.* proteins, nucleic acids. Exploit differences between the shape and other features of the chemical or biochemical nature of the molecules.

5.2 EQUILIBRIUM CONSIDERATIONS

Adsorption equilibrium is a dynamic concept achieved when the rate at which molecules adsorb on to a surface is equal to the rate at which desorb.

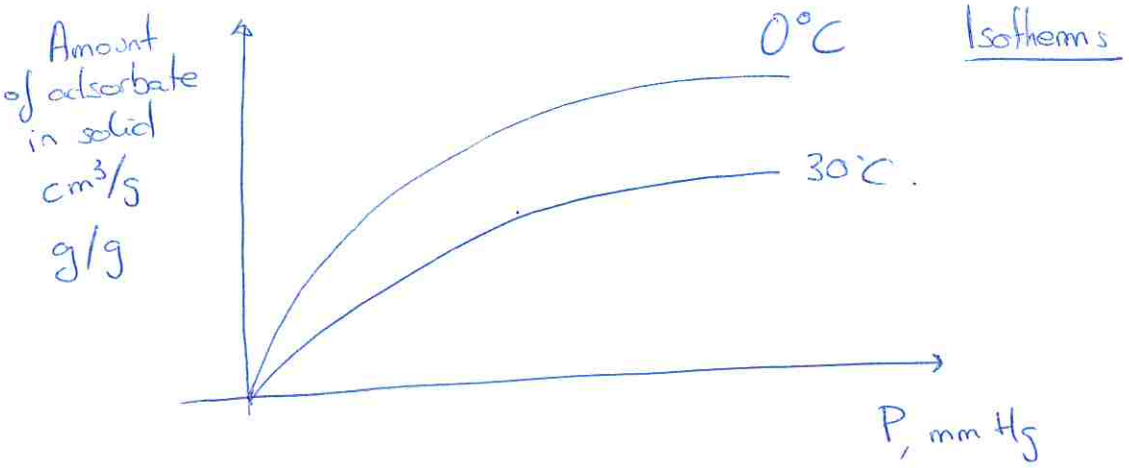
$r_{adsorption} = r_{desorption}$

The equilibrium partitioning of the adsorbate between the fluid and the solid phase can be studied experimentally.

The capacity of an adsorbent for a particular adsorbate depends on: the concentration of adsorbate in the fluid phase, the concentration of the adsorbate in the solid phase and the temperature of the system.

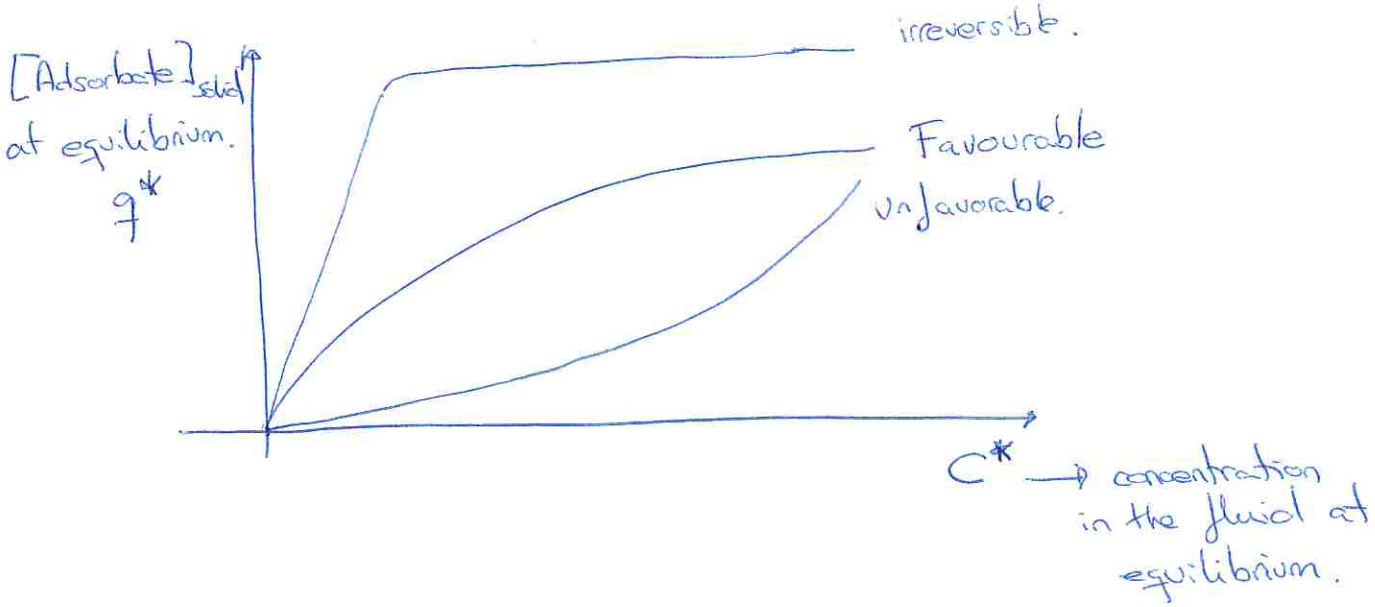
If one of these parameters are kept constant, the other two can be plotted to represent the equilibrium. It is common to keep the temperature constant and the result is an adsorption *isotherm*. If the concentration of the solute in the solid phase is kept constant, the data is known as an adsorption isotere.

Because adsorption is exothermic, the temperature significantly affects the amount of adsorbate adsorbed. Consequently, at a given pressure, the amount of adsorbate adsorbed at equilibrium decreases with increasing temperature.



The interaction between the adsorbent and adsorbate lead to different types of isotherms:

- Favourable isotherm: has a convex shape, representing a large amount of adsorption at a low partial pressure (or for liquids, concentration).
- Unfavourable isotherm: has a concave shape and a relatively high partial pressures (or concentration) is needed to achieve economic adsorption
- Irreversible isotherm: where maximum adsorption is reached at very low partial pressures (or concentrations).



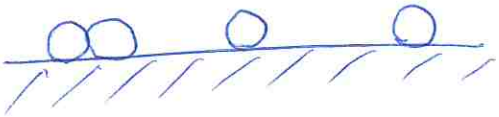
Note that an isotherm that is favourable for adsorption is unfavourable for desorption:

5.2.1 Langmuir Isotherm

Assume the adsorption of a single molecule, A, at the surface. The fraction of sites occupied and unoccupied are:

θ_A = fraction of occupied sites

$(1-\theta_A)$ = fraction of unoccupied sites.



Where:

$$\theta_A = \frac{n_{ads}}{n_m \rightarrow}$$

N_m: number of molecules of a monolayer

The adsorption rate of A is directly proportional to the partial pressure of A and to the fraction of free surface sites. This proportional factor is called adsorption constant k_a. Thus:

$$r_{ads} = k_a \cdot p_A \cdot (1-\theta_A)$$

(k_a = f(T))

However, simultaneously occurs desorption of adsorbed A on surface sites. The desorption rate of A is directly proportional to the fraction of occupied sites or adsorbed A molecules and k_d is the desorption constant:

$$r_{des} = k_d \cdot \theta_A$$

At the equilibrium, the rate of adsorption equals the rate of desorption which means that molecules are constantly adsorbed and desorbed allowing the adsorption of new molecules and free sites.

$$r_{ads} = r_{des}$$

$$k_a \cdot p_A (1-\theta_A) = k_d \cdot \theta_A$$

solve for θ_A

The fraction of occupied sites of A molecules are:

$$\theta_A = \frac{k_a p_A}{k_d + k_a p_A}$$

The equilibrium adsorption-desorption constant is a thermodynamic property defined as:

$$b = \frac{k_a}{k_d}$$

Thus:

$$\theta_A = \frac{b\,p_A}{1 + b\,p_A}$$

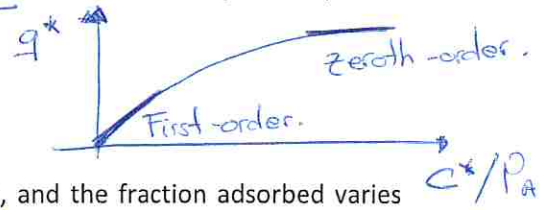
This is the Langmuir equation of a single adsorption molecule. Note that the fraction of surface sites can be determined experimentally by measuring adsorbed volumes in a system of constant pressure and temperature, according to the ideal gas law:

$$\theta_A = \frac{V_{ads}}{V_{monolayer}}$$

The order of magnitude of the equilibrium constant, *b* indicates the affinity of the gas with the solid. For high *b* values, the equilibrium is shifted towards adsorption and vice versa. There are two extreme cases to be considered:

b ↑ → high affinity
b ↓ → low affinity.

- If $b\,p_A \sim 0$, the equation becomes $\theta_A = bp_A$ where the surface coverage is proportional to the partial pressure of A. The situation occurs when $p_A \sim 0$ or on the initial stretch of the isotherm or when $b \sim 0$ when the gas affinity with the solid is very small.
- If $b\,p_A >> 0$, the equation becomes $\theta_A = 1$ or, the coverage is a complete monolayer. It occurs when $p_A >> 0$, near the vapour pressure of A or when $b >> 0$ when there is a high affinity of gas with the surface.



One can distinguish two regions:

- At **low partial pressures**, the denominator approaches 1, and the fraction adsorbed varies linearly with partial pressure in what is known as the first-order region or the Henry's law region. In this region, a higher partial pressure increases the tendency towards adsorption.
- As **pressure increases**, the number of available adsorption sites decreases, and further adsorption becomes increasingly difficult. At high pressures, the denominator in the Langmuir equation approaches $b\,p_A$, and the fraction of occupied sites increase asymptotically to 1. This is called the zero-order region.

For liquid-phase adsorption, the same procedure applies, but using concentrations:

$$r_{ads} = k_a \cdot C\,(1 - \theta_A)$$

$$r_{des} = k_d \cdot \theta_A$$

At equilibrium:

equivalent to b

$$\theta = \frac{k_a C^*}{k_d + k_a C^*} = \frac{C^*}{K_d + C^*} \text{ where } K_d \equiv \frac{k_d}{k_a}.$$

Here, C^* means the concentration in the liquid phase in equilibrium with the sorbent and it is customary to define a dissociation constant K_d for the adsorbed species.

$\theta = \frac{n_{ad}}{n_{monolayer}} \simeq \frac{V_{ad}}{V_{monolayer}}$

Now, the fractional coverage of the adsorption sites $\theta = \frac{q^*}{Q_m}$, → concentration of solute on solid at equilibrium

→ max capacity.

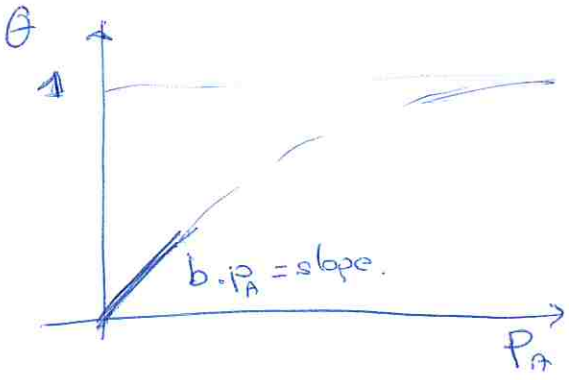
where:

- q^* is the amount of adsorbed adsorbate (per kg adsorbent), and
- Q_m is the maximum capacity of the adsorbent (per kg adsorbent).

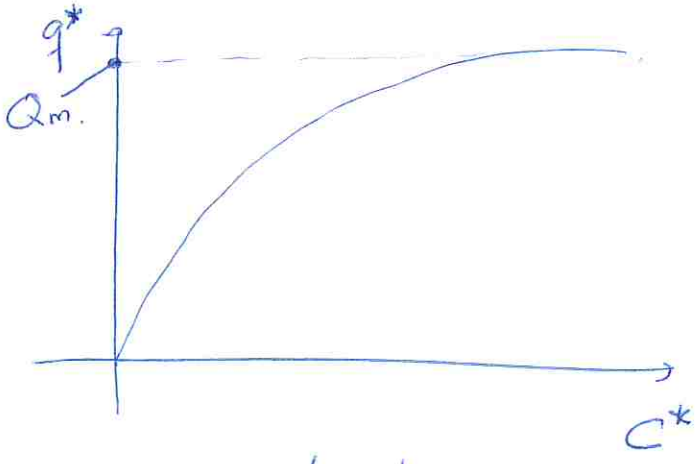
$$q^* = \frac{Q_m C^*}{K_d + C^*}$$

(Note in some situations, the amounts of adsorbed adsorbate are expressed per unit volume of adsorbent, as opposed to per unit mass).

The shape of the classic Langmuir isotherm is:



Gases



Liquid.

By analogy with the Monod equation and the Michaelis Menten equation, the two characteristic parameters K_d and Q_m can best be calculated from a reciprocal (straight-line) plot.

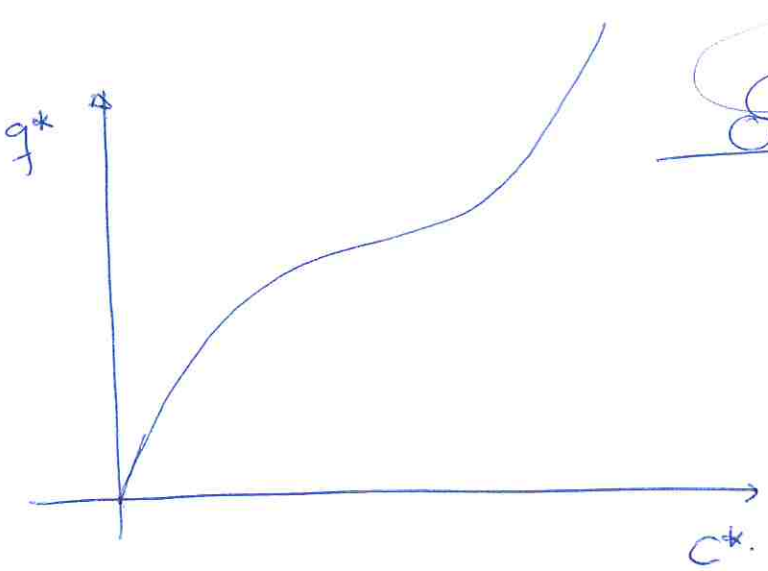
e.g.
$$\frac{1}{q^*} = \left(\frac{K_d}{Q_m}\right) \cdot \frac{1}{C^*} + \frac{1}{Q_m}$$

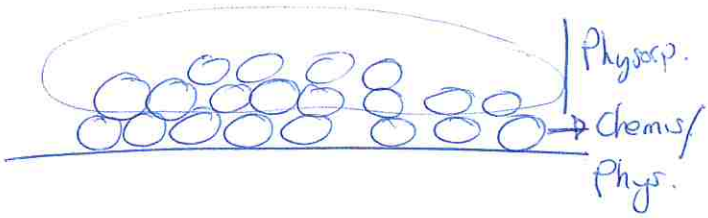
5.2.2 Freundlich Isotherm

A semi-empirical isotherm of the form:

$$q^* = b \cdot \left(C^*\right)^m$$

where b and m are constants and $m < 1$.



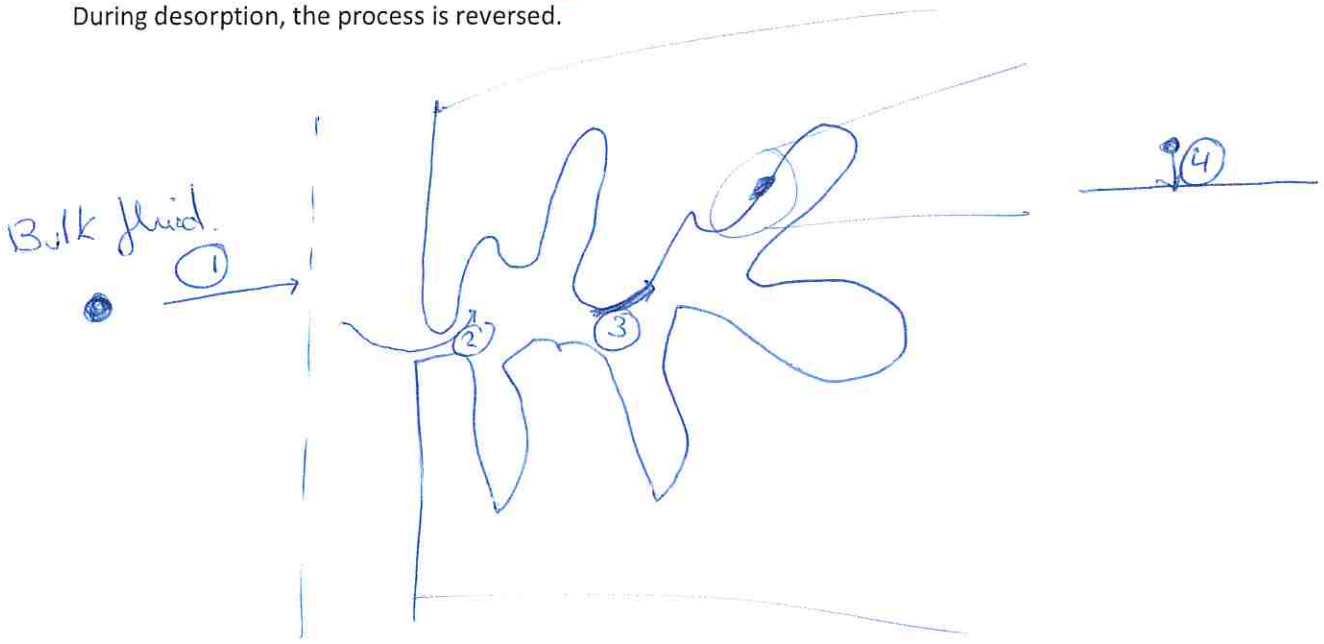


5.3 KINETICS AND TRANSPORT

The overall kinetics of adsorption can be affected by a number of different resistances to mass transfer:

- External film resistance in bulk fluid
- Diffusion along pores to an adsorption site (for a porous adsorbent)
- Surface diffusion of adsorbed molecules migrating along the walls of the pores
- The adsorption process itself.

During desorption, the process is reversed.



An accurate description of all of these different processes is difficult and is a non-steady state problem as adsorption sites become increasingly occupied as adsorption proceeds.

Approximations are therefore frequently used, for example in liquid phase systems:

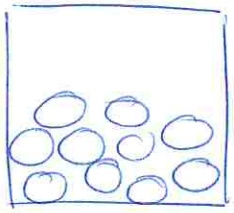
$$\text{Rate of mass transfer} = K_C\, a\, (C - C^*)$$

where

- K_C is an overall coefficient of mass transfer,
- C^* is the concentration in the bulk liquid phase that would be in equilibrium with q, the amount adsorbed on the solid phase.

5.4 GAS – PHASE ADSORPTION

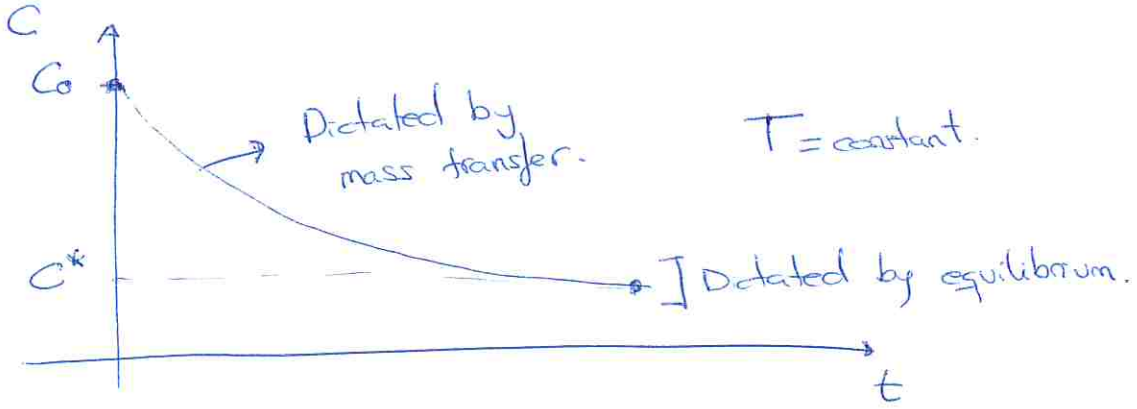
Unlike the situation for gas absorption, it is nearly impossible to operate a truly continuous counter-current system because of the problems involved with the continuous mechanical conveying of the solid phase (such as attrition and particle break up). [This is not an issue in gas/liquid or liquid/liquid absorption systems.]



5.4.1 Batch Operations

It consist of the following steps:

- Add adsorbent to an agitated tank of liquid feedstock
- Allow system to reach equilibrium: this is equivalent to one equilibrium stage.



- Adsorbent is recovered (*e.g.* by filtration) and regenerated.
- Stirring speed, particle size, adsorbent loading *etc.* will affect shape of uptake curve.

To find the equilibrium position, a material balance on the substrate at equilibrium gives:

$$V\,C_0 = V\,C^* + M\,q^*$$

where

- V is the volume of fluid in the batch
- C_0 is the original concentration of adsorbate at time $t = 0$
- M is the mass of adsorbent particles added to the batch.

$$q^* = \frac{V\left(C_0 - C^*\right)}{M}$$

At equilibrium (using, say, the Langmuir isotherm for liquid systems):

$$q^* = \frac{Q_m C^*}{K_d + C^*}$$

Langmuir isotherm.

Hence, substitute for q^* and solve the resulting quadratic in C^* (check that you have identified the appropriate root $0 < C^* < C_0$).

$$\frac{Q_m \cdot C^*}{K_d + C^*} = \frac{V(C_0 - C^*)}{M}$$

Use these design equations to determine:

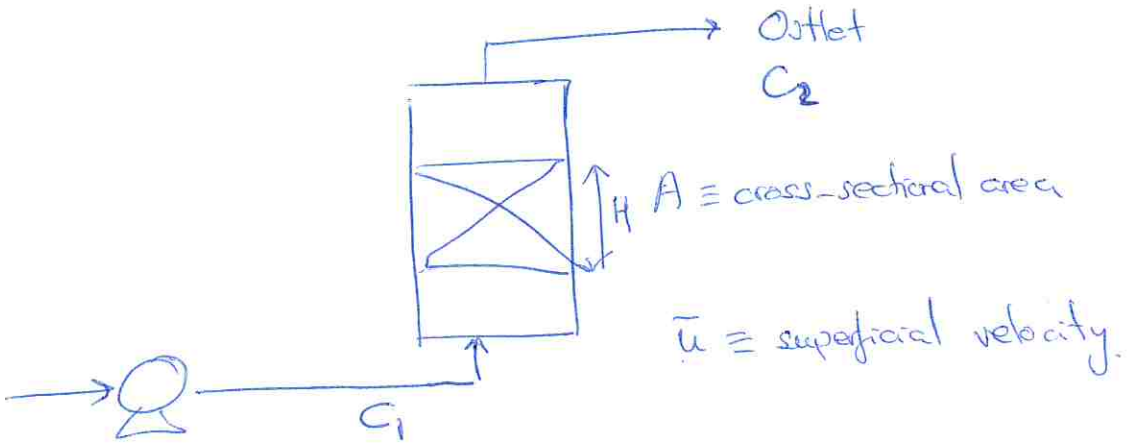
- the mass of adsorbent (M) that needs to be added to reduce C_0 to C^*
- the value of C^* that will be achieved at equilibrium when a mass M of adsorbent is added to a batch system.

Q1-2

5.4.2 Packed bed columns

Pack adsorbent into a fixed bed and flow feed continuously through it (inlet conc. C_1).

This is a non-steady state situation. The adsorption capacity of the bed gradually becomes exhausted and adsorbate is no longer efficiently adsorbed by the bed.

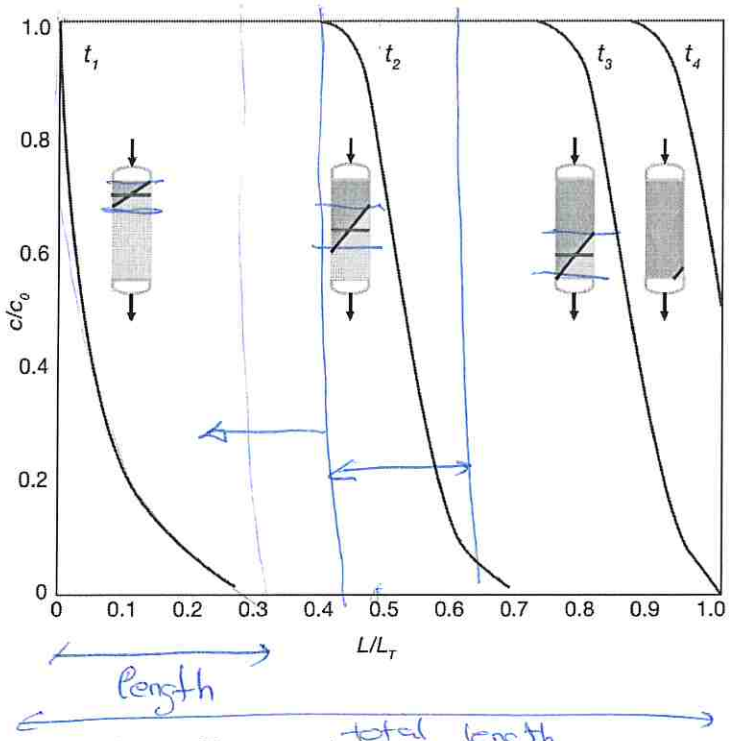


The concentrations in the fluid phase and solid phase change with time and position in the bed:

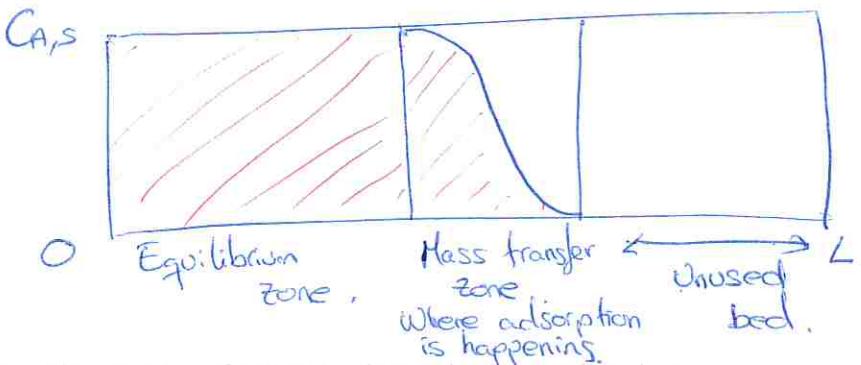
- Mass transfer starts immediately upon introduction of the feed causing the adsorbate concentration to decrease along the length of the bed until it reaches nearly zero.
- As fresh feed continues to enter the column, a portion of the bed reaches equilibrium with the feed, and no additional net mass transfer occurs in that part of the bed.
- When the feed reaches a section of the bed that has not yet achieved equilibrium, mass transfer resumes and the fluid concentration again decreases with increasing bed length, until

it reaches nearly zero. The length over which the concentration changes is called ***the mass-transfer zone***.

- As the process continues, more and more adsorbent reaches equilibrium with the feed. The feed must travel farther and farther to reach a portion of the bed that has not reached equilibrium. Thus, the mass-transfer zone progresses along the length of the bed and an increasing portion of the bed becomes used or inactive.



At a given time, the concentration profile across the bed is:

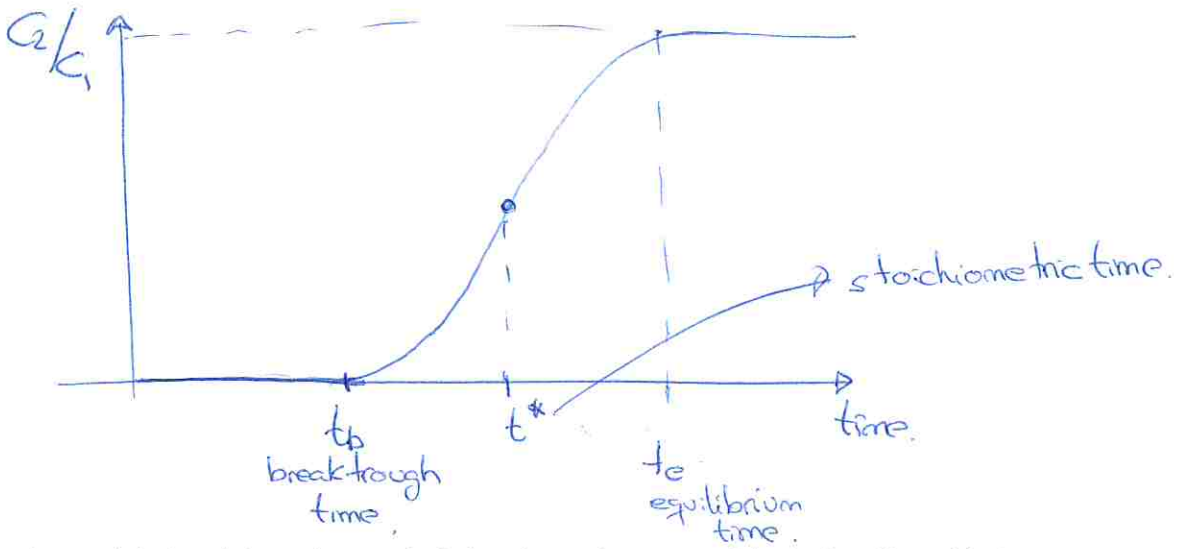


- The mass-transfer zone is where adsorption takes place.
- An S-shaped indicates there is mass transfer resistance and axial dispersion and mixing.

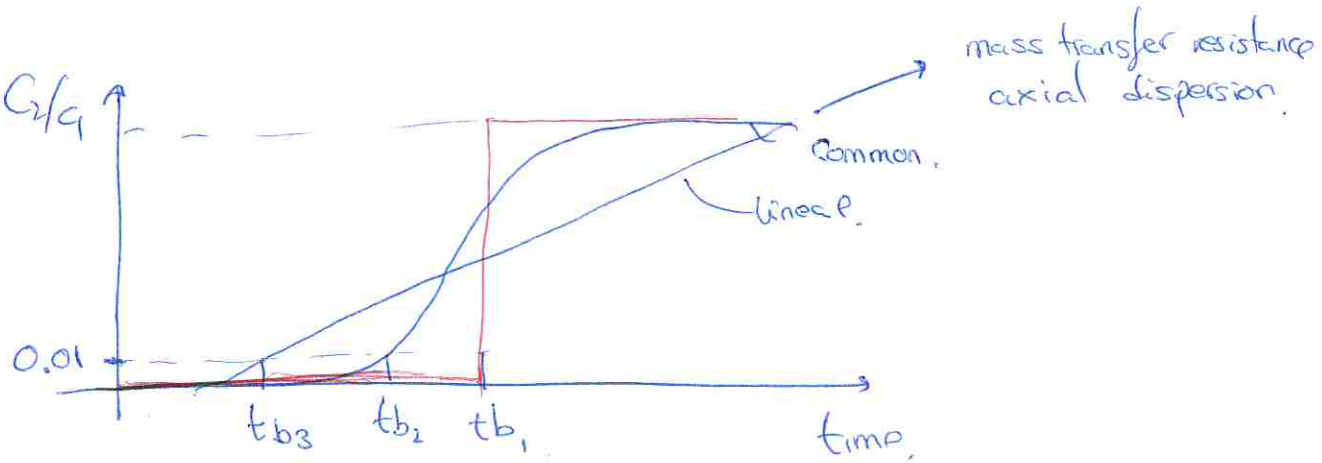
- An ideal case will result in a perfectly vertical line moving through the bed.
- The equilibrium zone is where the isotherm applies

At some point, the adsorbate leaves with the column effluent (i.e. it is not totally adsorbed in the bed). This is called **breakthrough time**.

Looking at what is happening at the exit from the bed, the following concentration profile will be seen. This is known as the **breakthrough curve** as adsorbate breaks through the bed and is not totally adsorbed in the bed:



The shape of the breakthrough curve is vital to the performance of the bed and is a critical consideration in the design of packed bed adsorption operations:

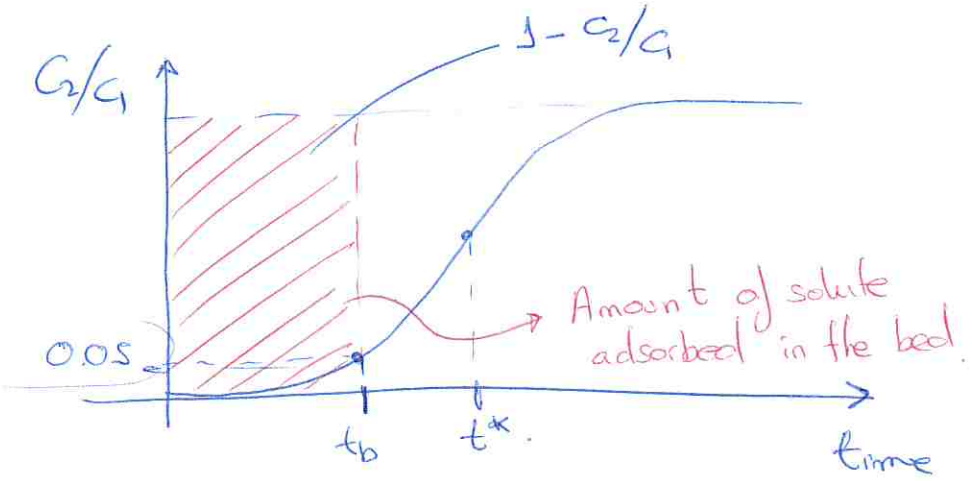


Factors that influence the sharpness of the breakthrough curve are:

- flowrate of fluid through the bed
- shape of equilibrium isotherm
- mass transfer rates in the bed and the adsorbent particles,
- size of bed.

The steeper the breakthrough curve, the longer is the useful service time of the bed i.e. the time until the level of adsorbate in the exit exceeds a desired value. Here $t_{b1} > t_{b2} > t_{b3}$.

The amount of adsorbate adsorbed is proportional to the area above the breakthrough curve:



The total amount adsorbed by the bed is then given by:

$$M\,q = u\,A\,C_1 \int_0^t \left(1 - \frac{C}{C_1}\right) dt$$

Max amount of solute in outlet stream that is allowed.

Where:

M is the mass of adsorbent in the packed bed,

q is the average capacity of the bed per unit mass of adsorbent,

A is the cross-sectional area of the bed,

u is the superficial velocity of the flow, and

the integral represents the area above the breakthrough curve, i.e. the fraction of the total amount of adsorbent that has been fed to the bed (uAC_1t), that has been captured by the adsorbent in the bed.

If the breakthrough curve has a symmetrical S-shape, the midpoint is known as stoichiometric time, or t^*. The portion of the bed that is equilibrated is the real breakthrough time divided by the stoichiometric time. Conversely, the portion of unused bed is:

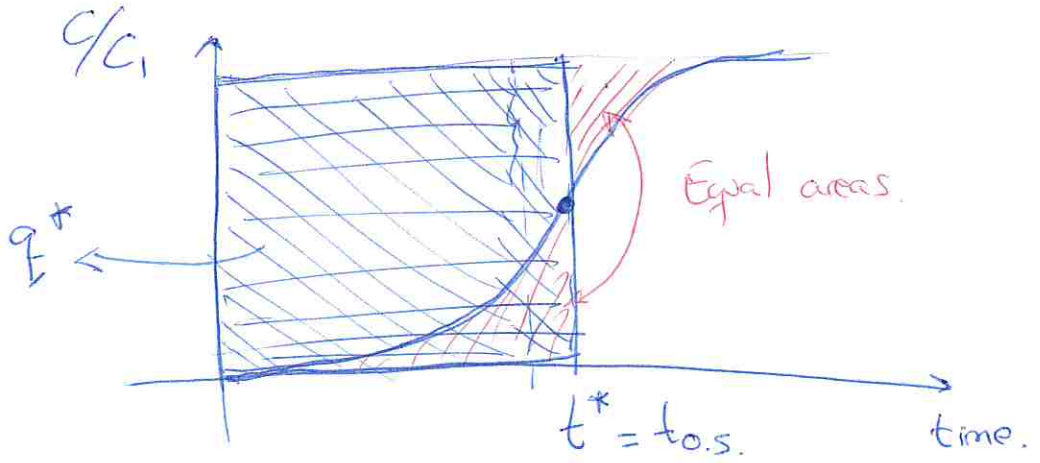
$$\frac{\text{length of unused bed at } t_b}{\text{Total length}} = 1 - \frac{t_b}{t^*}$$

If we assume that the mass-transfer zone moves through the bed unchanged, then, the length of unused bed at breakthrough (LUB) does not change with bed length. Consequently, bed utilisation efficiency improves with length, because the unused portion becomes a smaller fraction of the overall bed.

However, in a real column, the mass-transfer zone does change as it moves through the bed due to back mixing and axial dispersion.

The shape of the breakthrough curve is symmetrical about $C/C_1 = 0.5$, which occurs at break time $t_{0.5}$, then the saturation capacity of the bed, q^*, (which would be achieved after an infinite volume of feed has been supplied to the column) can be readily estimated:

$$M q^* = u A C_1 \int_0^\infty \left(1 - \frac{C}{C_1}\right) dt = u A C_1 t_{0.5}$$

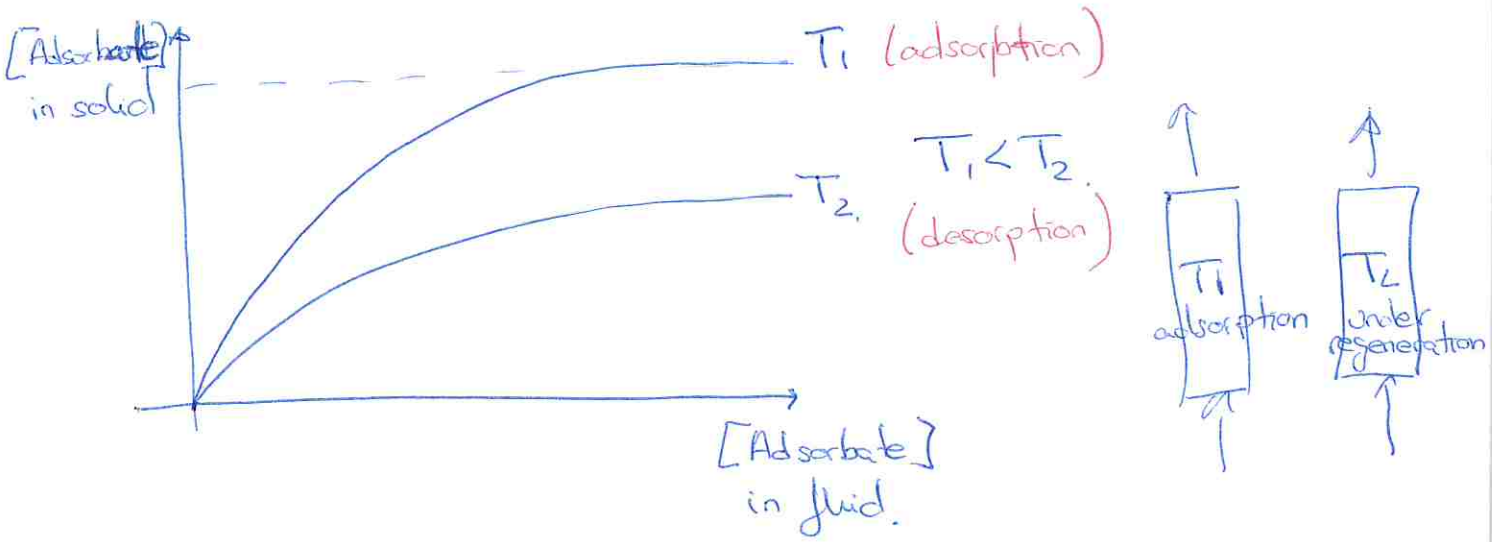


5.4.3 Regeneration of spent adsorbent

In the design of an adsorption unit, it is important to ensure that the spent adsorbent can be regenerated in a given time and that the total inventory of adsorbent is kept to a minimum.
The equilibrium condition has to be changed for the regeneration to occur. This is usually achieved by changing the temperature or pressure so that the driving force, which has previously promoted the adsorption of the adsorbate in the solid, is reversed.

Thermal swing

The simplest and most common way of regenerating an adsorbent is by heating.
The vapour pressure exerted by the adsorbed phase increases with temperature, so that molecules desorb until a new equilibrium with the fluid phase is established.

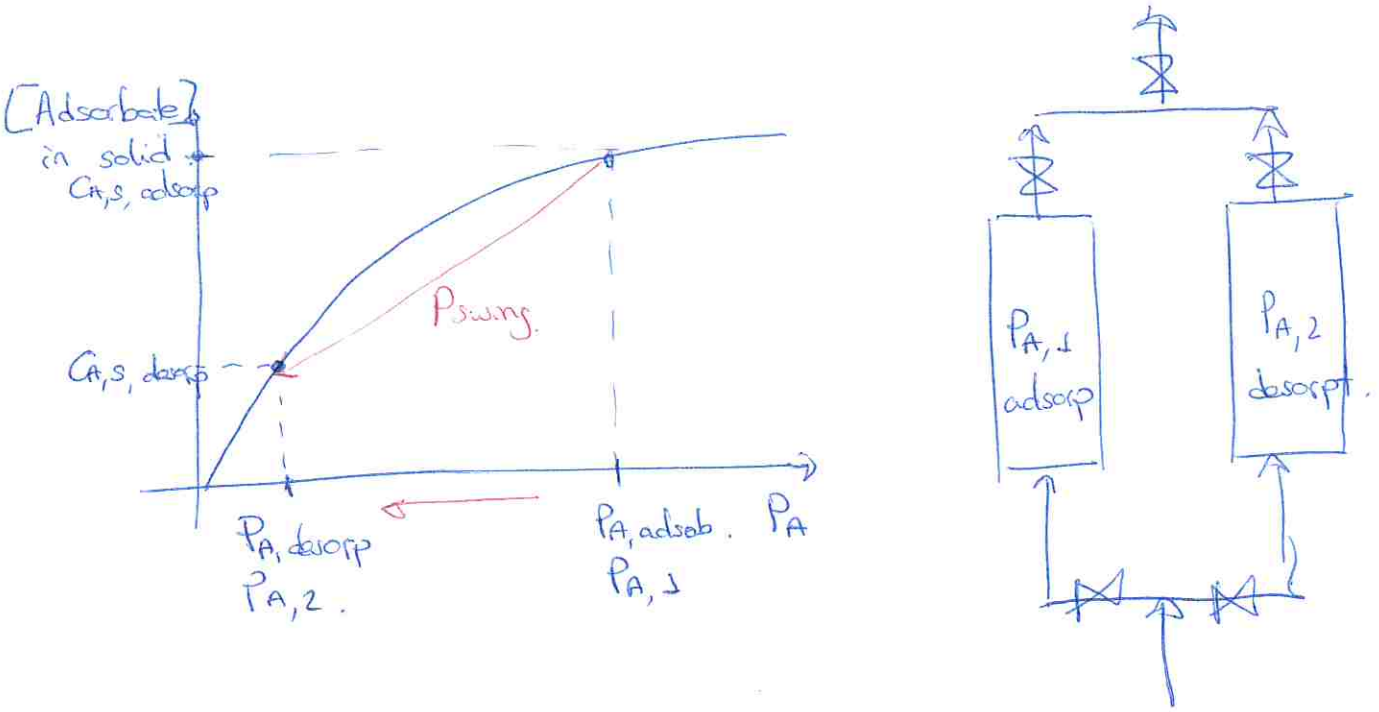


The adsorption unit using thermal swing regeneration usually consists of two packed beds, one on-line and one regenerating.
The adsorption temperature depends on the properties of the adsorbent (to avoid coalesce and capacity reduction) and the adsorbates.
The relatively poor conductivity of a packed bed makes it difficult to heat the bed quickly. Thermal regeneration is normally associated with long cycle times, measured in hours.

Pressure swing

An alternative way of regenerating the adsorber is by promoting adsorption at high pressures and desorption at low pressures, hence called pressure swing adsorption.

A typical arrangement consists of two fixed beds, one adsorbing and one regenerating, which are then, reversed.



A cycle consists of 4 steps:

1. High-pressure feed flows through bed A. Part of its effluent is expanded to the lower pressure and then passed through bed B which it regenerates.
2. Bed B is re-pressurised to the feed pressure using feed gas, while pressure in bed A is dropped to the purge pressure.
3. Steps 3 and 4 follow the same sequence than Steps 1 and 2 but the functions of beds A and B are reversed

Pressure-swing regeneration is particularly useful when the stream to be treated is needed at pressures above atmospheric.

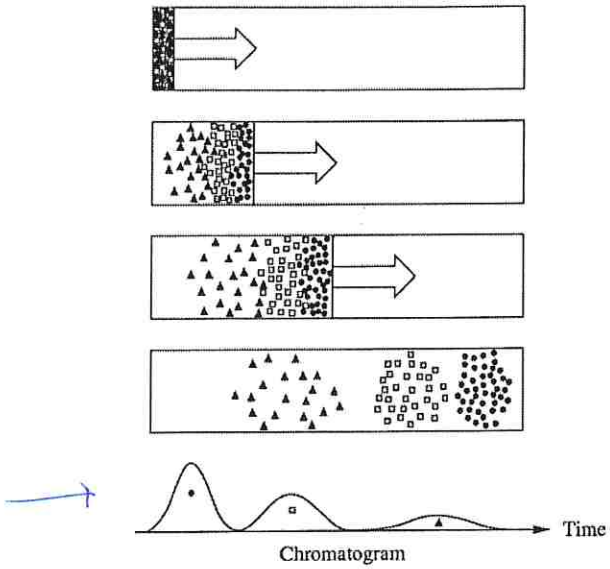
When the process stream is at atmospheric pressure or below, it might also be possible to regenerate the adsorbent using partial vacuum.

5.5 LIQUID-PHASE ADSORPTION

Most of the principles associated to gas-phase adsorption can be equally applied to liquid-phase adsorption when the main objective is primary isolation of species.

However, when the objective is the enrichment and purification of species in liquid-phase via adsorption, the processes are called **chromatography**.

Chromatography relies on the differential migration of species through a packed column with a resin using an eluent where different components move through the resin at different rates and they come out at **different elution times**, enabling the separation.



The resin is the **stationary phase** and the liquid that carries the solute is the **mobile phase.**

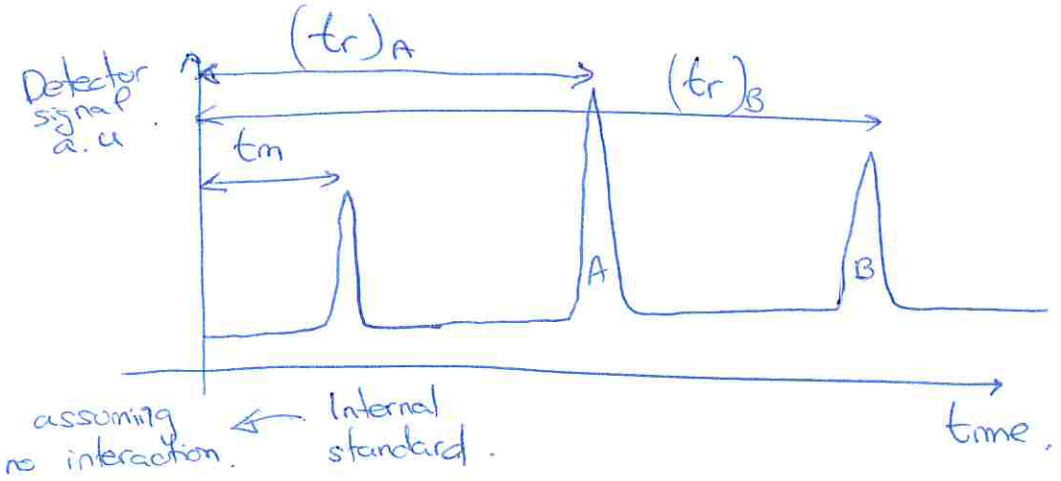
The graph of concentration of the various components in the effluent vs elution time is call **chromatogram**.

There are different types of liquid chromatography depending on the basis to generate differential migration, such as adsorption chromatography, partition chromatography, gel chromatography, ion-exchange chromatography and affinity chromatography.

You will study them in more detail in Bioprocessing Engineering.

Chromatographs are evaluated to perform qualitative and quantitative analysis of the species present in the liquid bulk phase.

- For qualitative analysis, **external standards** containing the eluted species are needed.
- For quantitative analysis, **internal standards** with known concentrations and elution times are usually recommended.



The area of each peak is proportional to solute concentration. The relationship between both is called **calibration line**.

The retention time an analyte spends in the system is composed of two components:

- T_m: is the time it takes a solute to pass through the space occupied by the mobile phase (dead space, column void volume). It has the same value for all the analytes.
- T'_r: the adjusted retention time represents the time the analyte spends retained by the stationary phase. It represents the separation process or interaction of the analyte with the stationary phase.

$$t_r' = t_r - t_m$$

Partition coefficient (also called distribution coefficient) - is the ratio of a compound's concentration in the stationary phase to its concentration in the mobile phase. It determines how a compound distributes itself and it is used for separation.

$$K = \frac{C_s}{C_m}$$

$K = 1 \Rightarrow$ equal distribution.

When K=1 the analyte is distributed equally

Capacity factor (also called partition ratio or capacity ratio) – is the ratio of the total amount of a solute in the stationary phase to the amount in the mobile phase at equilibrium.

$$k' = \frac{m_s}{m_m}$$

Its value can also be related to the elution time which is easier to measure:

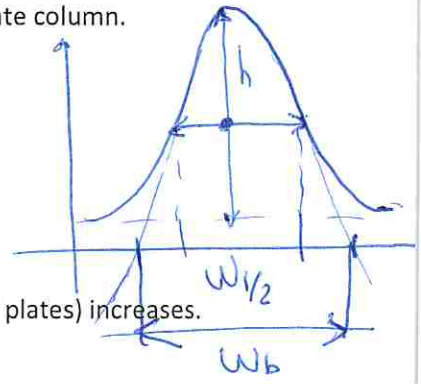
$$k' = \frac{t_r - t_m}{t_m} = \frac{t_r'}{t_m}$$

The efficiency of a column to separate analytes is related to the sharpness and shape of the peaks. Ideally, one aims for symmetrical, shape peaks.

Two general approaches have been developed to measure the column efficiency:

1. **Plate Theory:**

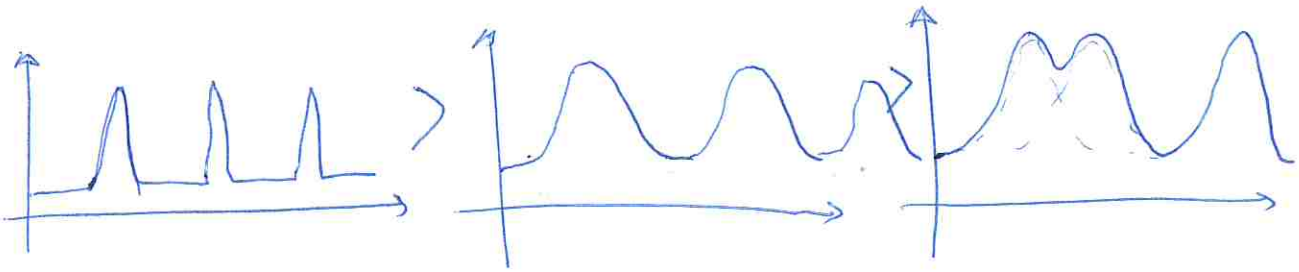
It models a chromatographic column as a series of narrow, discrete sections called theoretical plates. At each plate, the solute is assumed to reach an equilibrium between the mobile and stationary phases. According to plate theory, a column is mathematically equivalent to a distillation plate column.



The **number of theoretical plates** can be calculated by:

$$n = 5.54 \left(\frac{t_r}{w_{1/2}}\right)^2 = 16 \left(\frac{t_r}{w_b}\right)^2 \quad \text{(assuming a Gaussian peak)}$$

The efficiency of a column increases as the number of equilibrations (i.e. theoretical plates) increases. Analytes can have the same elution time but different number of plates:



Similarly, the **effective plate number** is calculated as:

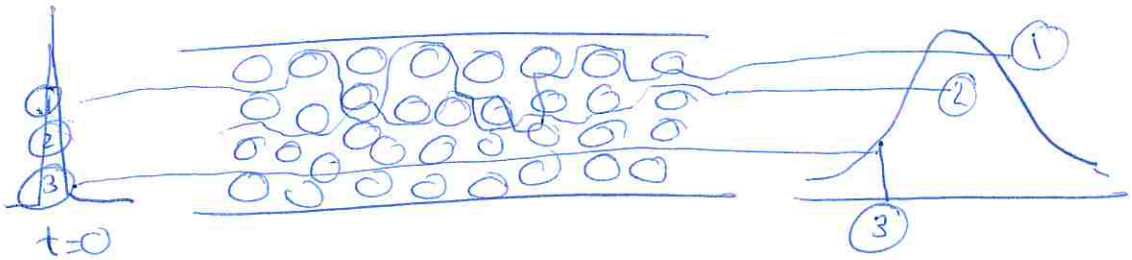
$$N = 5.54 \left(\frac{t_r - t_m}{w_{1/2}}\right)^2 = 5.54 \left(\frac{t'_r}{w_{1/2}}\right)^2 \text{ (for symmetrical peaks)}$$

$$N = \frac{41.71(t_r/w_{0.1})^2}{A_s + 1.25} \quad \text{(for asymmetrical peaks)}$$

where A_s is the asymmetry factor.

2. Rate theory

It explains peak broadening and peak separation by analysing the dynamic processes of solute molecules moving through the stationary phase.



Column Dispensivity, H, is assumed to be the sum of the individual contributions of the kinetic factors.

It uses the van Deemter equation:

$$H = \frac{A + B}{u + Cu}$$

Plate height. → velocity

Where u is the average linear mobile phase velocity

A: represents the contribution to zone broadening by eddy diffusion

B: represents the contribution of longitudinal diffusion

C: represents the contribution of resistance to mass transfer in both the stationary and mobile phases.

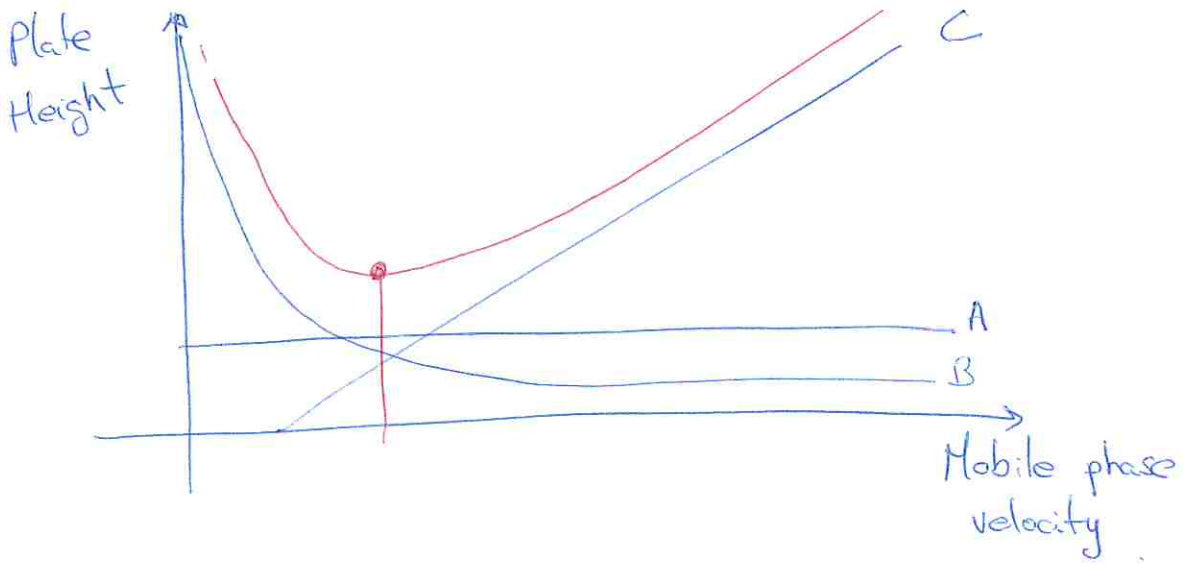
Each of these parameters can be calculated by correlations based on properties of the column.

TABLE 26-2 Variables That Influence Column Efficiency

Variable	Symbol	Usual Units
Linear velocity of mobile phase	u	cm s^{-1}
Diffusion coefficient in mobile phase*	D_M**	$cm^2 s^{-1}$
Diffusion coefficient in stationary phase*	D_S	$cm^2 s^{-1}$
Retention factor (Equation 26-12)	k	unitless
Diameter of packing particles	d_p	cm
Thickness of liquid coating on stationary phase	d_f	cm

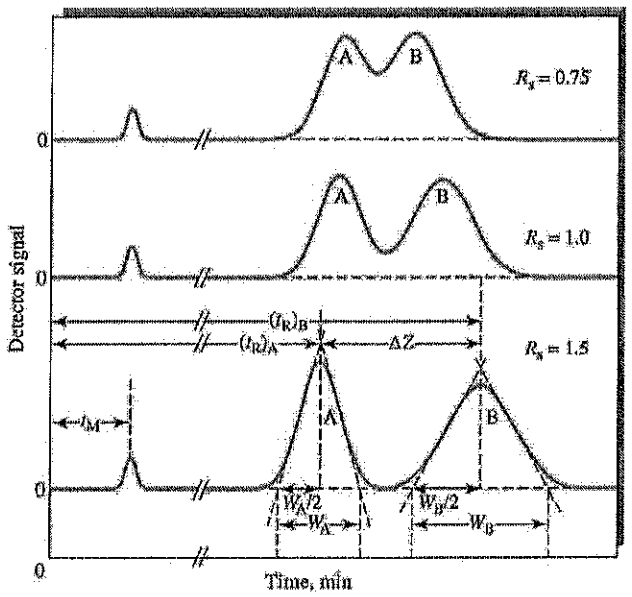
TABLE 26-3 Processes That Contribute to Band Broadening

Process	Term in Equation 26-23	Relationship to Column* and Analyte Properties
Multiple flow paths	A	$A = 2\lambda d_p$
Longitudinal diffusion	B/u	$\frac{B}{u} = \frac{2\gamma D_M}{u}$
Mass transfer to and from stationary phase	$C_S u$	$C_S u = \frac{f(k) d_f^2}{D_S} u$
Mass transfer in mobile phase	$C_M u$	$C_M u = \frac{f'(k) d_p^2}{D_M} u$



Knowing how well a column retains a solute is not sufficient as one normally needs to deal with multiple eluents.

Resolution (R_s) describes how well two neighbouring peaks are separated from each other.



The fundamental Resolution equation is:

$$R = \frac{1}{4}\sqrt{2}\ x\ \frac{k}{k+1}\ x\ (\alpha - 1)$$

Where: α is the selectivity (= k_2/k_1) – influenced by mobile and stationary phase

N is the column efficiency – influenced by length by length and particle size

K is the capacity factor (retention – influenced by the stationary and mobile phase